**10.** The curve $C$ has equation

$$x = 3\sec^2 2y \qquad x > 3 \qquad 0 < y < \frac{\pi}{4}$$

(a) Find $\frac{dx}{dy}$ in terms of $y$.

**(2)**

(b) Hence show that

$$\frac{dy}{dx} = \frac{p}{qx\sqrt{x-3}}$$

where $p$ is irrational and $q$ is an integer, stating the values of $p$ and $q$.

**(3)**

(c) Find the equation of the normal to $C$ at the point where $y = \frac{\pi}{12}$, giving your answer in the form $y = mx + c$, giving $m$ and $c$ as exact irrational numbers.

**(5)**